How this book is organized: A roadmap

The book has three parts that cover 12 chapters.

Part 1 (chapters 1-4) covers blockchain basics and the design and development of smart contracts.

Chapter 1 introduces the 3 Ds of blockchain--decentralization, disintermediation, and distributed immutable ledger--and provides a high-level conceptual view of a blockchain.

Chapter 2 is a gentle introduction to smart contracts on the Ethereum blockchain, applying design principles to develop smart contracts, code them with the Solidity language, deploy them in a web-based Remix integrated development environment, and transact with them. Smart contracts for a decentralized counter (Counter.sol) and an airline consortium (ASK.sol) are developed.

Chapter 3 is about techniques for adding trust and integrity to the smart contract code. A ballot smart contract (Ballot.sol) representing voting in a digital democracy is introduced and developed in incremental steps.

Chapter 4 introduces the design and development of a decentralized application (Dapp) with smart contract logic and a web-based user interface. A Node.js-based Truffle suite of tools is introduced for developing and running the smart contract and the web application. The Ballot